Asking for Evidence Helps, Enforcing It Can Hurt: Citation-Grounded Extraction of Airline Refund and Change Conditions

(A Pilot Benchmark)

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Abstract

Airline refund, cancellation and change conditions are published as fare rules and terms pages written in inconsistent semi-structured language, and travel agents still read them by hand. We release a pilot benchmark for extracting these conditions into a 12-field schema: 250 clauses from 12 public airline documents (11 carriers), 166 labelled with fields and 84 marked as stating no rule. We compare a keyword rule baseline, a local 7B model (Qwen2.5-7B) and two frontier models (Claude Sonnet 5.5 and Opus 5.5), each with and without a *citation* condition: the model quotes the clause text behind every value, and a gate removes values whose quote, or whose value within the quote, does not occur in the clause. The gate checks textual traceability, not whether the quote supports the value. Our main finding separates the two halves of this condition. Asking frontier models for quotes improves extraction by 3 to 5 F1 points; the gain holds against both annotators and the adjudicated labels, when whole documents rather than clauses are resampled, and in every pairing of three repeated runs, and the same request makes the models leave clauses without a rule empty more often. The gate itself moves their scores by less than one point but removes 18.6 points from the 7B model, which leaves half of its values without a quote. Absolute scores depend on the annotation: inter-annotator agreement is moderate (topic $\kappa = 0.51$); frontier models beat the rules against the primary annotator's labels but not against the second annotator's; and the best micro F1 (75.4, on clauses known to state a rule) is dominated by the topic and action fields. Over all 250 clauses, including those without a rule, the best systems reach 65 to 66 F1, and placed behind a retriever the best extractor recovers 45 to 51% of gold field values instead of 79%.

1 Introduction

When a passenger asks to change or refund a ticket, an agent must find what the fare allows: whether the change is permitted, what it costs, until when, and whether the fare difference applies. These conditions live in ATPCO fare-rule categories (Cat 16 penalties, Cat 31 voluntary changes, Cat 33 voluntary refunds) and in carriers' public conditions of carriage, written as free text with carrier-specific layouts. Misreading them costs money in both directions: unrecoverable penalties, or refunds that were never due.

Large language models can read this text, but an agent needs to know *where* an answer came from before acting on it. A common answer is to require the model to cite its source. We ask two questions:

1. How well do current models extract refund and change conditions into a fixed schema, compared with hand-written rules?

2. Does requiring citations help, and if so, is it the request for evidence or the enforcement of it that matters?

Contributions:

- **A pilot benchmark** of 250 airline policy clauses with a 12-field schema, a 123-line annotation guide, a second-annotator agreement study, an adjudicated subset and a blind self-relabel (Section 3).
- **A comparison** of rules, a local 7B model and two frontier models under plain, citation, chunk-context and guideline prompts (Section 4).
- **An ablation that separates citation prompting from citation enforcement** using each run's own pre-gate values (Section 5.2). For frontier models the gain comes from asking for evidence; the verbatim gate is nearly free for them and harmful for a small model.
- **An audit of label provenance**: some gold labels were revised in consistency passes drafted with the help of one of the evaluated model families. We report every result on the untouched labels as well (Section 3.4).
- **Checks on what the scores mean**: per-field results with their support, scoring of the clauses without a rule, document-level resampling, and a retrieval step in front of the extractor (Sections 5.4–5.8).

This is a pilot: the scored set is small (132 records after near-duplicate removal, from 12 documents), and the gold labels come mainly from one annotator. We treat the results as evidence about effect directions and failure modes, not as a leaderboard; the citation finding is the result that holds under every check we ran.

2 Related work

Extraction from contracts and policies. Legal and policy text has established extraction benchmarks: CUAD labels 41 clause categories in 510 commercial contracts, annotated by trained law students (Hendrycks et al., 2021), and OPP-115 annotates data practices in 115 website privacy policies (Wilson et al., 2016). Both show that expert-defined schemas and careful annotation, not model choice alone, determine what can be measured. Airline conditions are closer to policy text than to contracts, but carry typed values (fees, currencies, percentages, time windows) that span annotation does not capture; we label typed fields instead.

Structured extraction with LLMs. Large language models extract schema-shaped records from technical text with little or no task-specific training (Dagdelen et al., 2024, for materials chemistry). We use prompted frontier and open models without fine-tuning and score typed fields against human labels.

Attribution and citation. Rashkin et al. (2023) define attribution to identified sources (AIS) as a standard for checking generated statements against evidence, and ALCE benchmarks LLMs that generate text with citations, finding that even strong models leave many statements unsupported (Gao et al., 2023). These works evaluate citations produced alongside free-text answers. We study a narrower setting, verbatim quotes for structured fields checked mechanically against the source clause, and separate the effect of asking for evidence from the effect of enforcing it.

Language technology for aviation text. Aviation NLP has concentrated on operational messages: NOTAMs have been processed with self-supervised language models for criticality

prediction, entity recognition and structured translation (Arnold et al., 2022), and recent work parses them with LLMs into structured, knowledge-grounded outputs (Liu et al., 2025, 2026). LLM agents have also been used to explain and resolve air traffic conflicts in simulation (Andriuškevičius and Sun, 2024). We did not find a public benchmark for extracting airline commercial conditions such as refund and change rules; this work targets that gap, on the passenger and agent side of aviation rather than the operational side.

3 Data

3.1 Sources and clauses

Of 27 public documents collected [`data/pilot_source_manifest.csv`], the labelling queue draws on 12 from 11 carriers: nine terms or conditions pages (Nok Air DD (two), Air Arabia G9, Malaysia Airlines MH, Qatar Airways QR, SpiceJet SG, Singapore Airlines SQ, Thai Airways TG, Oman Air WY), one fare-rule page (VietJet VZ), one policy bulletin (AirAsia AK) and one FAQ (Emirates EK). The other 15 are not in the benchmark: 7 travel-trade fare circulars were excluded as non-public trade documents; 7 public pages (six IndiGo fee tabs and one Bangkok Airways page) need a headless browser to render and yielded no parsed text; one Emirates page refused automated access and no saved copy was made. Snapshot dates recorded in the manifest are 2026-02-22/23 (VZ, DD, WY, G9, AK, SG) and 2026-10-28 (SQ, MH, QR, EK, TG); conditions change over time, so every label refers to its snapshot.

Documents were chunked and split into atomic clauses. Fare-rule pages are tables flattened to text ("Changes in Flight: Charge fee of change and difference in fare, if any"), so the splitter first mines recurring field labels from the corpus and splits only at those labels; other text is split at sentence and list boundaries. The labelling queue holds 250 clauses: 177 from terms pages, 53 from fare rules, 20 from policy bulletins. For analysis we group the six carriers whose pages present conditions as fee tables or fare-rule rows (VZ, DD, G9, AK, SG, QR) as *fare rule* and the rest as *policy text*.

Sampling. The 250 queue clauses were drawn from a pool of 3,578 clauses in three ways (Table 1). The first labelling round (62 clauses) covered the VietJet fare-rule page in full and 9 Nok Air clauses. Later rounds drew 181 clauses uniformly at random within each document, after removing fragments under 20 characters, failed splits over 700 characters, website chrome and exact duplicates. Because the random draw from the Singapore Airlines page returned only boilerplate, we added its 7 clauses that state a refund or cancellation rule by hand; this purposive addition is reported separately where it matters (Section 5.8). Scoring removes near-duplicates (same normalised text, one contained in the other, or word overlap of at least 0.85), which mostly affects VietJet, whose fare-rule rows repeat across fare brands (46 labelled clauses, 24 scored). One document thus supplies 18% of the scored clauses, which is why we also resample whole documents (Section 5.6).

3.2 Schema and annotation

Each clause gets either *skip* (no rule stated: menus, contact details, fragments) or 12 fields: policy topic (refund, cancellation, reissue/change, no-show, go-show, waiver, other), action (allowed, conditional, not allowed, unknown), penalty type (free, fare difference only, mixed, fixed amount, percentage, forfeiture, tax only, unknown), penalty amount, currency, percentage lost, fare difference required, time window, before/after departure, tax refund rule, eligibility conditions and exceptions. The guide [`docs/LABELING_GUIDE.md`] is built on two rules: label only what the clause states, and an empty field is a correct answer when the clause is silent. Topic follows the passenger's transaction, not the words used, following the ATPCO split between Cat 31 (changes, including residual value) and Cat 33 (refunds).

Carrier	Document	Pool	Queue	Labelled	Skipped	Scored	Selection
VietJet VZ	fare rules	53	53	46	7	24	first round (all)
AirAsia AK	policy bulletin	67	20	12	8	10	random
Nok Air DD	terms (1)	18	13	7	6	6	9 first round, 4 random
Nok Air DD	terms (2)	44	29	23	6	18	random
Air Arabia G9	terms	139	25	17	8	13	random
Qatar Airways QR	terms	427	25	14	11	14	random
SpiceJet SG	terms	319	8	3	5	3	random
Emirates EK	FAQ	109	11	3	8	3	random
Malaysia Airlines MH	terms	1,279	17	11	6	11	random
Singapore Airlines SQ	terms	16	10	7	3	7	3 random, 7 purposive
Thai Airways TG	terms	108	13	9	4	9	random
Oman Air WY	terms	999	26	14	12	14	random
Total		3,578	250	166	84	132	

Table 1: Clauses per document: in the clause pool, in the labelling queue, labelled or skipped by annotator A, and scored after near-duplicate removal.

Annotator A (the author) labelled all 250 clauses in a web tool with guardrails (field locking, format normalisation, pre-save warnings): 166 labelled, 84 skipped. The first 62 queue positions were labelled with the rule baseline's guess pre-filled; the rest were labelled blind. Section 3.3 measures what the pre-fill did. One guide amendment was made during the agreement study (2026-11-01: fixed fee plus percentage is `mixed`); it changed no existing gold label and was given to the second annotator in writing before they started.

3.3 Agreement

Inter-annotator. A second annotator (B) labelled a stratified sample of 50 clauses (42 labelled by A, 8 skipped; seed fixed) in a spreadsheet, alone, without AI tools and with only the guide; B worked with a colleague's help. B never saw A's labels or any model output [`results/v1c_iaa_summary_2026-11-01.md`]. Table 2 gives the agreement.

	Agreement	Cohen's κ
Label vs skip ($n = 50$)	90%	0.50
Topic ($n = 42$)	64%	0.51
Action ($n = 42$)	62%	0.44
All fields, micro F1	0.52	

Table 2: Inter-annotator agreement between A and B.

Disagreements were systematic: B labelled fare-rule table cells as `other` rather than `reissue_change`, used `unknown` where the guide leaves the penalty type empty, and labelled 5 clauses A skipped.

Adjudication. A and B resolved all 111 field disagreements (39 clauses) together, naming for each the guide rule that decided it [`results/v1c_adjudication.json`]. A's value was kept in 94, B's in 16, and a new value chosen once; no case was marked as a gap in the guide. Because the adjudication mostly kept A's values, we treat it as a consistency check on A, not as independent evidence.

Intra-annotator. A relabelled 27 of their own records blind, after the consistency passes [`results/v1c_self_relabel_summary_2026-11-02.md`]. Ten of these had been seen during adjudication the same day; on the other 17, topic κ is 0.82, action κ 0.60 and field micro F1 0.80 (all

27: 0.88, 0.65, 0.84). A is therefore much more consistent with themself than with B; the gap is between annotators' readings of the guide, and action is the least stable field for both.

Pre-fill anchoring. The 27 relabelled records are exactly the labelled records from the pre-filled part of the queue, so the blind relabel also measures anchoring. The rule baseline scores 47.9 F1 against the original (pre-filled) labels and 41.5 against the blind relabel (51.9 vs 43.1 on the 17 records not seen in adjudication), while the LLM systems move by up to 7 points in either direction with no consistent sign [`results/v1c_self_relabel.json`]. Pre-filling inflated the baseline on these records; it does not favour the LLM systems.

3.4 Label provenance

After first entry, the gold went through consistency passes in which fixes were drafted with Claude's help and approved by A. Since Claude models are evaluated, we compared database snapshots taken before each pass [`results/v1c_label_provenance_summary_2026-10-29.md`]. 68 of 250 records changed after first entry; of the 132 scored records, 100 are untouched. Claude models score higher on touched records than untouched ones (Opus + guide 84.5 vs 67.6) while the rule baseline scores lower (43.0 vs 50.7). We therefore report the untouched subset next to the full set. Snapshots record edits, not advice given while a label was first entered, so the untouched subset is a lower bound on contamination, not proof of its absence.

4 Methods

Rule baseline. Keyword and pattern rules mapping clause text to topic, action, penalty type and numeric fields.

LLM extraction. One request per clause with the schema and the rule "never guess; null when the clause does not state it"; every model gets the same prompt text for a given condition. Qwen2.5-7B-Instruct (the default `qwen2.5:7b` build in the Ollama library) runs locally on CPU through Ollama with JSON output mode, temperature 0 and a 4,096-token context. Claude Sonnet 5.5 (`claude-sonnet-5-5`) and Opus 5.5 (`claude-opus-5-5`) run through the Message Batches API at low effort, without constrained decoding. Replies are parsed as JSON and coerced to the schema (enumerations validated, numbers parsed); an unparseable reply counts as an empty prediction. Conditions:

- *plain*: clause text only;
- *+ cite*: the model must also return, for every non-null field, a verbatim quote from the clause. A local check normalises whitespace and case and keeps a value only if its quote occurs in the clause and, for number and text fields, the value occurs in the quote as a whole number or whole words (so 20 does not match "120"); list items must occur in the clause. Other values are set to null (the *gate*). The gate establishes textual traceability: every surviving value can be found in the clause. It does not check that the quote means what the value says (Section 5.7). The model's pre-gate values are stored, which makes the ablation in Section 5.2 possible;
- *+ ctx*: the parent chunk is shown as context, with values to be taken from the clause only;
- *+ guide*: a 22-line summary of the annotation guide [`docs/prompt_guidelines.txt`] is added to the prompt.

Not every combination was run for every model (Table 3), so comparisons between models use matched conditions only.

	plain	+ guide	+ cite	+ cite + ctx	+ cite + guide	repeated runs
Qwen 7B	✓	✓	✓	✓	✓	
Sonnet 5.5	✓		✓	✓		plain, + cite (3 each)
Opus 5.5	✓	✓	✓	✓	✓	plain, + cite (3 each)

Table 3: Conditions run per model. Qwen + cite + ctx (26.3 F1 [20.6, 31.5]) is left out of Table 4 because it does not differ from Qwen + cite.

Scoring. For each field present in gold, a prediction is correct if it matches (categorical: exact; numbers: equal; text: token F1 ≥ 0.5; time windows: equal durations). Two of these rules are lenient. Token overlap does not guarantee the same meaning for free-text conditions. And the gold time window records only the duration (the guide asks for it), so "within 24 hours" and "at least 24 hours" both match a gold value of 24 hours; the direction of a deadline is not scored. Predictions on fields that are empty in gold count as false positives. We report micro precision, recall and F1 over all fields, on gold with near-duplicate clauses removed (132 records: 88 fare rule, 44 policy text). Confidence intervals are 95% bootstrap intervals over records (1,000 resamples); differences between systems use a paired bootstrap (2,000 resamples), and we call a difference significant (marked *) when its interval excludes zero. Section 5.6 repeats the key comparisons resampling whole documents.

5 Results

5.1 Main results

Table 4 gives micro F1 (×100) against annotator A on the 132 scored records [`results/v1c_*.json`, view `all_dedup`]; the untouched column is from the provenance audit.

Method	P	R	F1 [95% CI]	Topic acc	Action acc	F1 untouched ($n = 100$)
Rule baseline	54.3	43.9	48.6 [44.0, 52.8]	73.5	51.5	50.7
Qwen 7B	52.1	39.9	45.2 [39.1, 50.7]	37.9	28.8	42.5
Qwen 7B + guide	62.6	45.3	52.5 [45.7, 58.0]	43.9	40.2	48.8
Qwen 7B + cite	49.0	17.3	25.6 [20.4, 30.7]	9.8	6.8	25.4
Qwen 7B + cite + guide	64.3	24.8	35.8 [28.6, 42.2]	23.5	18.2	27.4
Sonnet 5.5	57.5	70.9	63.5 [59.1, 67.8]	84.8	51.5	62.0
Sonnet 5.5 + cite	72.6	63.3	67.6 [63.0, 71.8]	80.3	45.5	65.5
Sonnet 5.5 + cite + ctx	73.1	68.0	70.5 [66.2, 74.9]	82.6	56.1	69.6
Opus 5.5	57.9	75.7	65.6 [61.4, 69.7]	87.1	64.4	62.4
Opus 5.5 + cite	67.3	72.3	69.7 [65.3, 74.1]	86.4	59.8	66.5
Opus 5.5 + cite + ctx	68.7	72.5	70.5 [65.9, 75.0]	87.9	59.8	67.7
Opus 5.5 + guide	67.0	78.8	72.5 [67.9, 76.8]	87.9	78.0	67.6
Opus 5.5 + cite + guide	72.7	78.2	**75.4** [71.1, 79.1]	87.1	79.5	72.6

Table 4: Main results against annotator A (×100, 132 records).

Paired differences (A's labels; full set / untouched): Opus over rules +17.1* / +11.8* [+5.1, +18.4]; Opus over Qwen +20.5* / +19.9*; Opus over Sonnet +2.1 / +0.4 (not significant); guidelines for Opus +6.8* / +5.2*.

Frontier models beat the rule baseline by a wide margin on A's labels, and the margin shrinks but remains significant on untouched labels; it does not hold against the second annotator's labels (Section 5.3). The 7B model is no better than rules with a plain prompt; the guide summary

raises it to 52.5 on all records but not on untouched ones (48.8, below rules at 50.7), so we make no claim for it. Sonnet and Opus are not distinguishable.

5.2 Citation: asking versus enforcing

"+ cite" bundles a prompt change (quote every value) and a filter (the gate). Each cited run stores the model's values before the gate, so we can score the same run with the gate off and split the effect exactly (Table 5) [`results/v1c_error_analysis_summary_2026-11-02.md`].

Model	Citation prompt (cite, gate off) − plain	Gate (on − off)
Qwen 7B	−0.9 [−3.7, +1.6]	**−18.6** [−23.8, −13.6]
Sonnet 5.5	**+4.4** [+1.8, +7.0]	−0.3 [−1.0, +0.3]
Opus 5.5	**+3.9** [+0.9, +6.6]	+0.2 [+0.0, +0.7]
Opus 5.5 + guide	**+2.8** [+0.4, +5.4]	+0.1 [+0.0, +0.3]

Table 5: Splitting the citation condition into the prompt change and the gate (F1 ×100, paired bootstrap 95% intervals, one run per condition).

For the frontier models, the improvement comes from asking for evidence: precision rises (Opus 57.9 to 67.3) because the model asserts fewer values it cannot point to. The gate then removes little, mostly eligibility list items, and the right and wrong values it removes roughly cancel (Opus + cite + guide: 33 right, 29 wrong or spurious). For the 7B model, the prompt change is neutral and the gate does the damage: Qwen returned a citations object but left 159 of its 303 asserted values without a quote; a further 42 had a quote that did not contain the value, 10 were quoted in the wrong format, and only 6 quotes were absent from the clause. The 7B model fails the gate by not following the output instructions, not by inventing quotes. Sonnet and Opus quote nearly every value; the values the gate removes for them (70 and 94) are mostly paraphrased eligibility items that do not occur word for word in their own quote.

Run-to-run variance. Table 5 uses one run per condition. To check that the prompt effect is larger than run-to-run noise, we repeated the four plain and cited Claude conditions twice more with identical settings, giving three runs each [`results/v1c_variance.json`]. Table 6 gives micro F1 as mean, standard deviation and range over the three runs.

	Sonnet 5.5	Opus 5.5
Plain	64.9, sd 1.4 [63.5, 66.3]	64.7, sd 1.1 [63.5, 65.6]
Cite, gate off	68.1, sd 0.2 [67.9, 68.4]	69.8, sd 0.4 [69.5, 70.2]
Cite	68.2, sd 0.5 [67.6, 68.6]	69.9, sd 0.2 [69.7, 70.2]
Prompt effect, all 9 pairings	**+3.3** [+1.7, +4.9]	**+5.1** [+3.9, +6.7]
Gate effect, per run	+0.1 [−0.3, +0.4]	+0.2 [+0.0, +0.3]

Table 6: Three runs per condition (F1 ×100, 132 records). Effects are mean [min, max].

Every pairing of a cited run (gate off) with a plain run is positive for both models. The nine pairings reuse three runs per condition, so they show the spread of the effect rather than nine independent replications. The size of the effect varies with the runs paired: the single-run estimates in Table 5 (+4.4 for Sonnet, +3.9 for Opus) lie within these ranges, and the three-run means are the better estimates. The gate effect stays within half a point in every run. Plain runs vary more than cited runs (sd 1.1 to 1.4 against 0.2 to 0.5).

5.3 Dependence on the annotator

We rescored all systems on the 42 clauses both annotators labelled, against A, against B and against the adjudicated labels (Table 7) [`results/v1c_iaa_summary_2026-11-01.md`; `results/v1c_error_analysis_summary_2026-11-02.md`, section 3].

Claim	vs A	vs B	vs adjudicated
Citation prompt, Sonnet	+4.7*	+5.3*	+4.9*
Citation prompt, Opus	+5.7*	+5.9*	+4.5*
Gate, Qwen 7B	−19.1*	−13.0*	−18.3*
Gate, Opus	0.0	0.0	0.0
Opus − rules	+18.7*	+0.8	+21.3*
Opus + cite + guide − rules	+25.5*	+7.7	+27.0*
Opus − Qwen 7B	+20.4*	+9.5*	+21.5*

Table 7: Each claim rescored against each label set on the 42 doubly labelled clauses (F1 ×100; * significant).

The citation findings hold against every label set. The advantage of frontier models over rules does not hold against B alone: against B, no LLM condition is significantly better than rules. Models agree with A (0.70 to 0.76 F1) more than B does (0.52). This is consistent with the provenance audit and with B applying some guide rules differently (Section 3.3). We therefore present the LLM-over-rules result as conditional on the annotation and the citation result as the robust finding.

5.4 What the micro F1 measures

Topic and action are filled in every scored record, so they make up 264 of the 444 gold values and dominate micro F1. The monetary fields are rare: 4 penalty amounts, 4 currencies and 1 percentage (Table 8) [`results/v1c_robustness.json`]. Their scores rest on so few cases that we draw no conclusion from them, and the benchmark does not yet test the financial details that motivated it. We add three summaries fixed before computing them: macro F1 over the seven fields with at least 10 gold values; micro F1 over the ten value fields (everything except topic and action); and whole-record accuracy, the share of records with all 12 fields right and nothing spurious.

Opus + cite + guide stays best on every summary, but the summaries show how much the headline hides. The rule baseline's micro F1 of 48.6 comes mostly from topic and action: on the value fields it scores 21.8, below the 7B model (46.5). The best system gets all 12 fields of a record right a third of the time. Free-text eligibility conditions, the largest value field, reach at most 53.4 for any system.

5.5 Clauses without a rule

The scores so far use only the clauses A labelled. A deployed extractor also sees the 84 clauses A skipped as stating no rule, where every extracted value is wrong. We ran each Claude condition on these clauses with the same prompts and settings (Qwen runs are pending) and scored all clauses together: 132 with a rule and 75 without, after removing near-duplicates [`results/v1c_rule_detection.json`]. A system *predicts a rule* when it fills any field; we fixed this definition before the runs.

Two results stand out (Table 9). First, asking for quotes makes the models leave no-rule clauses empty more often (Sonnet from 32% to 57%, Opus from 15% to 27%), which is consistent with the precision gain in Section 5.2. Second, the guide condition does the opposite: with it,

Field	Gold	Rules	Qwen 7B	Sonnet + cite	Opus	Opus + cite + guide
Policy topic	132	73.5	49.8	83.1	87.5	87.5
Action	132	51.5	38.0	58.5	67.5	79.8
Eligibility conditions	62	18.9	34.4	50.0	43.4	53.4
Penalty type	44	20.0	33.3	61.0	66.7	76.6
Before/after departure	22	46.7	45.7	63.6	37.5	52.5
Time window	19	13.3	86.5	80.0	72.0	76.6
Fare difference required	15	23.5	93.3	100.0	96.6	96.6
Tax refund rule	6	25.0	50.0	40.0	50.0	40.0
Penalty amount	4	20.0	60.0	72.7	66.7	80.0
Penalty currency	4	20.0	46.2	46.2	33.3	37.5
Exceptions	3	20.0	0.0	40.0	30.8	40.0
Penalty percentage	1	0.0	66.7	100.0	100.0	100.0
Micro F1 (all fields)	444	48.6	45.2	67.6	65.6	75.4
Macro F1 (7 fields, $\geq$ 10 gold)		35.4	54.4	70.9	67.3	74.7
Micro F1 (value fields)	180	21.8	46.5	62.0	53.4	64.3
Whole-record accuracy	132	9.1	6.1	12.1	12.1	33.3

Table 8: F1 per field (×100) with the number of gold values, and three summaries, against annotator A on the 132 scored records.

	Rule F1	Specificity	Values on no-rule clauses	F1, rule clauses only	F1, all clauses [document CI]
Rule baseline	77.9	0.0	158	48.6	40.6 [35.6, 45.5]
Sonnet 5.5	83.8	32.0	69	63.5	59.4 [54.8, 63.8]
Sonnet 5.5 + cite	86.5	57.3	34	67.6	65.0 [59.5, 70.8]
Sonnet 5.5 + cite + ctx	85.1	40.0	53	70.5	66.4 [61.1, 70.8]
Opus 5.5	80.5	14.7	179	65.6	55.9 [49.1, 61.6]
Opus 5.5 + cite	82.8	26.7	77	69.7	64.3 [57.0, 70.0]
Opus 5.5 + guide	78.6	4.0	185	72.5	60.8 [50.3, 69.9]
Opus 5.5 + cite + ctx	83.8	32.0	71	70.5	65.4 [57.9, 71.2]
Opus 5.5 + cite + guide	79.0	6.7	149	75.4	64.9 [55.5, 72.7]

Table 9: Rule detection and field F1 including the clauses without a rule (×100; 132 clauses with a rule, 75 without). Specificity is the share of no-rule clauses left empty.

Opus fills a field on 93 to 96% of no-rule clauses, and over all clauses the best system on rule clauses (75.4) falls to 64.9, level with Sonnet + cite + ctx (66.4) and Opus + cite + ctx (65.4). Most of these values are topic `other` with action `unknown` (59 of the 84 clauses for Opus + cite + guide), which reads as "nothing here" in schema terms. Counting that pattern as no rule, a definition we chose after seeing the outputs, raises the guide-prompted specificity to 39% (Opus + guide) and 73% (Opus + cite + guide). Either way, the ranking of the best systems depends on whether clauses without a rule are scored, and the gain from the guide summary is partly a cost on those clauses.

5.6 Document-level robustness

The 132 scored clauses come from 12 documents, and clauses of one document share wording, so intervals from resampling clauses can be too narrow for new documents. Table 10 repeats the key comparisons with a document-level (cluster) bootstrap and gives the range of each difference when one document at a time is left out [`results/v1c_robustness.json`].

Document-level intervals are wider, as expected, but every claim keeps its sign and significance. The citation-prompt effect for Opus + guide is borderline on all records (lower bound +0.04) and clear on untouched records. No single document drives any of the effects. This addresses dependence within documents only; with 12 documents from 11 carriers, generalisation to other

Claim	Δ	Document CI	Leave one out	Untouched: Δ [document CI]
Citation prompt, Sonnet	+4.4	[+1.9, +6.9]*	[+3.5, +5.0]	+4.1 [+1.0, +7.6]*
Citation prompt, Opus	+3.9	[+0.8, +6.3]*	[+2.9, +4.5]	+3.7 [+0.6, +6.9]*
Citation prompt, Opus + guide	+2.8	[+0.04, +6.1]*	[+1.8, +3.8]	+4.9 [+1.9, +8.0]*
Gate, Qwen 7B	−18.6	[−26.3, −8.9]*	[−20.2, −15.5]	−16.2 [−25.5, −8.1]*
Gate, Sonnet	−0.3	[−1.4, +0.5]	[−0.6, 0.0]	−0.6 [−2.1, +0.5]
Gate, Opus	+0.2	[0.0, +0.7]	[0.0, +0.3]	+0.3 [0.0, +0.9]
Opus − rules	+17.1	[+8.7, +26.9]*	[+12.5, +18.9]	+11.8 [+3.9, +18.8]*
Opus + cite + guide − rules	+26.8	[+16.4, +36.3]*	[+22.6, +29.4]	+21.9 [+13.3, +29.8]*
Opus − Qwen 7B	+20.5	[+13.5, +31.0]*	[+18.9, +23.0]	+19.9 [+9.0, +31.4]*

Table 10: Key differences in F1 (×100) against annotator A, with document-level bootstrap intervals (2,000 resamples of the 12 documents) and the range over leaving out each document.

carriers remains untested, and these comparisons are all against annotator A.

5.7 Error analysis

For the best system (Opus + cite + guide) [`results/v1c_error_analysis.json`]:

1. **Permission with a condition read as "unknown"** (17 of 26 action errors, mostly policy text). Clauses that state a fee without saying the change is allowed ("Passenger must pay service fee THB 225 per passenger") are `conditional` in the guide; the model treats them as not stating permission. Action is also the least reliable field for humans (intra κ 0.60, inter 0.44).
2. **Values the clause does not state.** Before/after departure is filled when gold is empty 18 times, time window 9, currency 8, often from amounts in fee tables that are not penalties (e.g. a paid refund add-on).
3. **Change versus other.** Name-change fees and upgrade terms that mention change fees are labelled `other` (5).
4. **Eligibility conditions.** Free-text lists are the noisiest field to score (16 wrong, 15 missed) and the one the gate removes most.

A worked example. Figure 1 shows the pipeline, and Table 11 one clause from the benchmark as Opus + cite processed it. Every quote occurs in the clause, so the gate keeps every value. Four of the five values are right. The fifth restates the whole rule as an eligibility condition; it is quoted verbatim and passes the gate, but the gold leaves the field empty, so it is scored as spurious. The time window shows the leniency of the scorer: the prediction says "at least 24 hours", the gold records 24 hours, and the match ignores the direction.

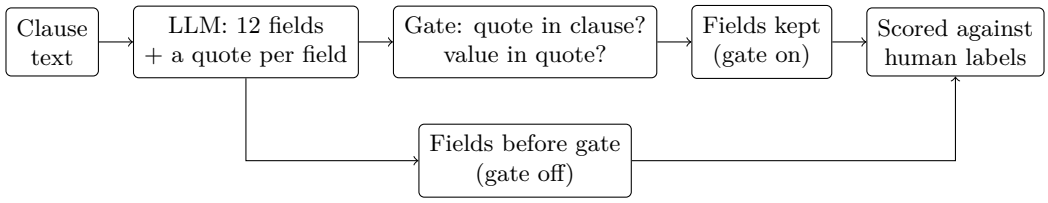


Figure 1: The citation condition. The model's values are stored before the gate, so each cited run can be scored with the gate on and off.

Clause: "The flight change must be made at least 24 hours prior to the original departure time."			
Field	Gold	Prediction (all quotes found in the clause)	Scored
Policy topic	reissue/change	reissue/change	correct
Action	conditional	conditional	correct
Before/after	before departure	before departure	correct
Time window	24 hours	"at least 24 hours prior to the original departure time"	correct (duration only)
Eligibility	(empty)	"The flight change must be made at least 24 hours prior to the original departure time"	spurious

Table 11: One clause, Opus + cite. The gate keeps all five values; one is wrong.

5.8 Retrieval before extraction

All results above give the system the clause that holds the answer. A deployed system must first find it. We built one query per document and topic for which annotator A labelled at least one clause with that topic (23 queries, 83 known-relevant clauses; topic `other` has no query). Each topic has one fixed query wording, written before any result was seen (e.g. "Can the flight or travel date be changed, and what change fee applies?"). Within each document we rank either all its clauses (3,179 after removing exact duplicates) or the passages they were split from (575 chunks; a clause counts as found when its chunk is retrieved, as in a retrieve-then-extract pipeline), with BM25, a dense encoder (`bge-small-en-v1.5`) and their reciprocal-rank fusion [`results/v1c_retrieval.json`].

Only 7% of the clause pool was sampled for labelling, so precision cannot be measured: an unlabelled retrieved clause may or may not be relevant. We report recall of the known-relevant clauses; because the labelled clauses were sampled uniformly within each document (apart from the first 62 queue positions and 7 purposive Singapore Airlines clauses), this estimates recall over all relevant clauses, and we check it on the uniformly sampled clauses alone. A random ordering gives the floor.

Unit	k	Random	BM25	Dense	Hybrid
Clause	5	11.5	28.9	31.3	31.3
	10	22.9	38.6	44.6	43.4
	20	41.2	56.6	63.9	63.9
Chunk	1	11.2	21.7	19.3	24.1
	3	33.7	49.4	42.2	42.2
	5	48.7	65.1	55.4	59.0

Table 12: Recall@k ($\times 100$) of the 83 known-relevant clauses, 23 queries. Random is the expected recall of a random ordering.

Retrieval loses many relevant clauses (Table 12). The intervals are wide (clause hybrid at $k = 10$: 43.4 [27.9, 58.8]; chunk BM25 at $k = 5$: 65.1 [44.2, 79.7]) and do not separate the three retrievers. Two kinds of clause are hard to find. Many relevant clauses are fragments of a longer rule ("the payment according to this channel dues immediately at change…"), which share few words with any query; retrieving chunks instead of clauses helps them. And long documents bury them: in the longest (1,161 clauses), four of its five relevant clauses are not in the hybrid

clause ranking's top 50, and in the second longest (945) both relevant clauses rank below 35. On the uniformly sampled clauses alone, recall is lower in every cell of Table 12, by up to 18 points (chunk BM25 at $k = 5$: 53.4), so the v1 and purposive records were easier to find than a typical clause. Even with the best retriever, 30% of queries have none of their relevant clauses in the top five chunks. Among retrieved clauses that A had judged, 64 to 100% were relevant at $k \leq 10$, but most retrieved clauses were never judged.

End to end, an extraction counts only if its clause was retrieved. Given the right clause, Opus + cite + guide recovers 78.8% of the gold field values of these 83 clauses (rules: 42.1%). Behind a retriever it recovers 30.5 to 37.0% with ten clauses and 45.3 to 51.4% with five chunks, depending on the retriever. Retrieval costs 27 to 48 points of field recall, comparable to the 37-point gap between the rule baseline and the best LLM given the right clause.

6 Discussion

Ask for evidence, but check the format before enforcing it. For capable models, a citation request is a cheap precision gain, and it also makes them leave clauses without a rule empty more often. A verbatim gate then adds traceability at almost no cost: every surviving value can be found in the clause. It does not add correctness, since a real quote can sit next to a wrong value (Table 11); checking that a quote entails the value would need a semantic test that we did not build. For the 7B model the gate deletes most of the output, but the cause is the output format: half of its values come without any quote. This shows that enforcing a format the model does not follow is harmful, not that small models cannot extract with evidence. Constrained decoding, re-asking for a missing quote, or flagging instead of deleting might avoid the collapse; we did not test them.

Score the whole task. The best configuration on clauses known to state a rule (Opus + cite + guide) loses its lead once clauses without a rule are scored, because the guide summary makes the model fill fields on almost every clause. Benchmarks of this kind should score rule detection and extraction together, or a system can look best by never abstaining.

Agreement is the bottleneck. Moderate human agreement (topic $\kappa = 0.51$) caps what any system score means. The disagreements cluster in a few guide decisions: how to treat fare-table cells, whether an unstated penalty type is `unknown` or empty, and when a stated fee makes an action `conditional`. Those are the guide changes to make before scaling the dataset.

7 Limitations

- Small scale: 132 scored records from 12 documents and 11 carriers, English only. Confidence intervals are wide, and several differences (Opus vs Sonnet, ctx and guide combinations on the 42-record subset) are not resolved.
- One primary annotator, who is also the author; the second annotation covers 50 clauses and was produced with a colleague's help; adjudication mostly kept the author's values. No annotator was a fare-rules specialist, and the guide was amended once during the agreement study.
- Some gold labels were revised in passes drafted with help from Claude, one of the evaluated model families. We report untouched-label results, but the audit cannot detect influence at first entry.
- The schema's monetary fields are almost untested (4 amounts, 4 currencies, 1 percentage), and tiered penalties and refund payout options, which the guide describes, are not scored.

- The scorer ignores the direction of a deadline ("within" versus "at least"), and token overlap of at least 0.5 can accept a free-text condition with a different meaning. The gate checks textual traceability only; no sample of accepted citations was checked by a person for whether the quote supports the value.
- Three runs per condition only for the plain and cited Claude conditions (Section 5.2); all other conditions, including Qwen and the guide and context variants, have one run. No variation of prompt wording was tested, and not all conditions were run for every model (Table 3). Qwen was not run on the clauses without a rule.
- Retrieval is evaluated only for recall, with 23 queries and one fixed wording per topic, on labels that cover 7% of each document; precision and answer quality after retrieval are not measured.
- Clause segmentation is fixed and imperfect; some clauses are fragments of tables.

8 Conclusion

On a pilot benchmark of airline refund and change conditions, asking frontier models to quote their evidence improves extraction by three to five F1 points, and the gain holds against both annotators, under document-level resampling and across repeated runs. Enforcing verbatim quotes costs those models nothing measurable and breaks a 7B model that does not supply quotes in the required format. Other conclusions are weaker: frontier models beat rules against the primary annotator's labels but not the second annotator's, the headline score is dominated by two classification fields, the best configuration loses its lead when clauses without a rule are scored, and behind a retriever the best extractor loses a third to more than half of its recall. An independently annotated evaluation set, with more monetary conditions, is the next step. The data, guide, agreement study, predictions, retriever rankings and scripts are released to support it.

Reproducibility

Code and data: `https://github.com/IhsanKabir/policy-extraction-rag`. The `release/` folder holds the clauses, all label sets (including the pre-fix snapshots used by the provenance audit), the samples, every system's predictions and the retriever rankings. `python scripts/reproduce.py` rebuilds the working files from it alone, reruns the scorer and every analysis script on the saved predictions, and checks every results file against the committed one; bootstrap seeds are fixed, so the check is exact. This reproduces the metrics, not the model outputs: rerunning inference uses `extractors/` with the same prompts and settings, but hosted models can change behind a fixed name, so new outputs may differ. `scripts/regate_check.py` re-applies the citation gate to every stored model reply and confirms the stored predictions. Dependency versions are pinned in `requirements-paper.txt`. All Claude runs together cost USD 11.28 through the Message Batches API (5.01 for the main runs, 4.18 for the repeats, 2.09 for the clauses without a rule).

Data statement. The clauses come from pages airlines publish so that passengers can read the conditions before buying. Copyright in the text remains with the carriers. We release the labels, source URLs, snapshot dates and clause text for research use, with attribution, and will remove any carrier's text on request. Code is released under MIT and the annotations under CC BY 4.0; the clause text is not covered by either licence.

Use of AI assistance. Claude (Anthropic) was used to write analysis code, to draft consistency fixes to the gold labels that the annotator then approved or rejected (Section 3.4), and to draft this paper. All labels were made or approved by human annotators, the second annotation and adjudication used no AI tools, and every reported number comes from committed scripts.

References

J. Andriuškevičius and J. Sun. Automatic control with human-like reasoning: exploring language model embodied air traffic agents. arXiv:2409.09717, 2024.

A. Arnold, F. Ernez, C. Kobus, and M.-C. Martin. Knowledge extraction from aeronautical messages (NOTAMs) with self-supervised language models for aircraft pilots. In *Proceedings of NAACL 2022: Industry Track*, pages 188–196, 2022.

J. Dagdelen, A. Dunn, S. Lee, N. Walker, A. S. Rosen, G. Ceder, K. A. Persson, and A. Jain. Structured information extraction from scientific text with large language models. *Nature Communications*, 15:1418, 2024.

T. Gao, H. Yen, J. Yu, and D. Chen. Enabling large language models to generate text with citations. In *Proceedings of EMNLP 2023*, 2023.

D. Hendrycks, C. Burns, A. Chen, and S. Ball. CUAD: an expert-annotated NLP dataset for legal contract review. In *NeurIPS 2021 Datasets and Benchmarks Track*, 2021.

M. Liu, Q. Fang, Y. Yang, C. Zhao, and K. Cai. Knots: a large-scale multi-agent enhanced expert-annotated dataset and LLM prompt optimization for NOTAM semantic parsing. arXiv:2511.12630, 2025.

M. Liu, Q. Fang, Y. Wu, C. Zhao, Y. Yang, and K. Cai. NOTAM-Evolve: a knowledge-guided self-evolving optimization framework with LLMs for NOTAM interpretation. In *Proceedings of AAAI 2026*, 2026.

H. Rashkin et al. Measuring attribution in natural language generation models. *Computational Linguistics*, 49(4):777–840, 2023.

S. Wilson et al. The creation and analysis of a website privacy policy corpus. In *Proceedings of ACL 2016*, 2016.